

Chapter-4

4.1) a)

Truth table

A	B	C	D	Output/F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

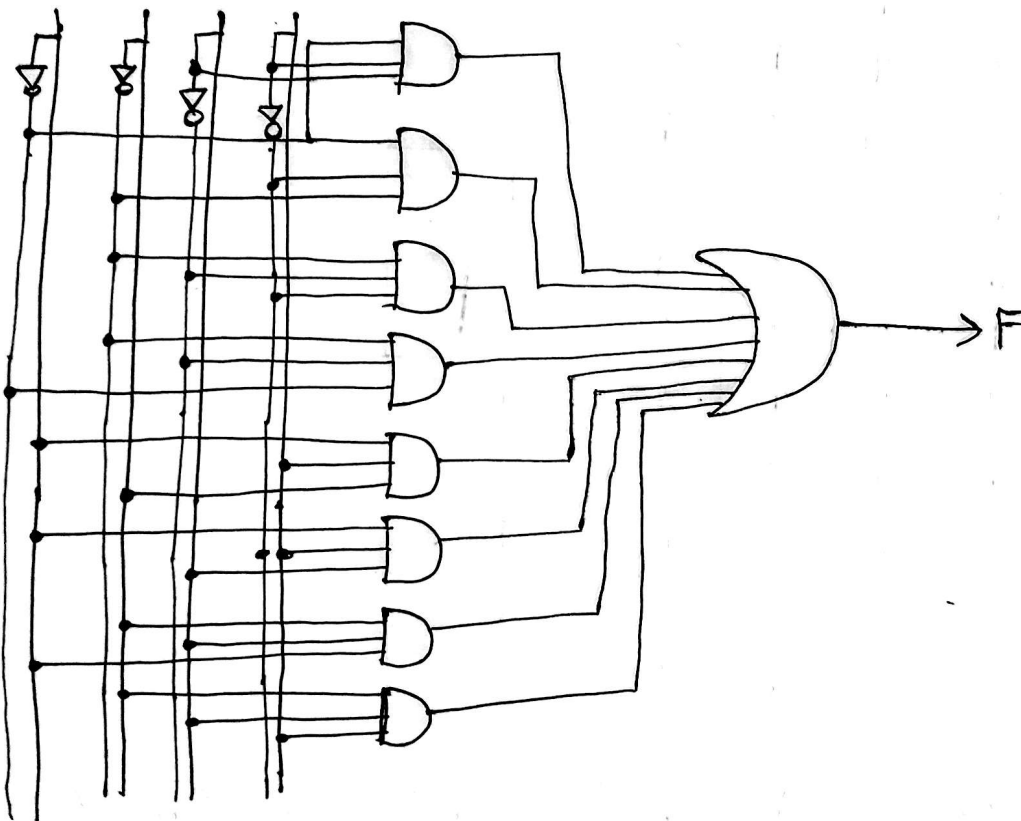
b) Sum of Product, $F = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D}$
 $+ \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + AB\bar{C}\bar{D} + ABC\bar{D} + ABCD$
 $\therefore F = \bar{A}\bar{D}\bar{C} + \bar{A}\bar{D}B + B\bar{C}\bar{D} + \bar{B}\bar{C}\bar{A} + ADB + ADC + BCA$
 $+ BCD$ (Ans)

c) Product of sums,.

$$F = (A+B+\bar{C}+\bar{D})(A+\bar{B}+C+\bar{D})(A+\bar{B}+\bar{C}+D) \\ (\bar{A}+B+C+\bar{D})(\bar{A}+B+\bar{C}+D)(\bar{A}+\bar{B}+C+D)$$

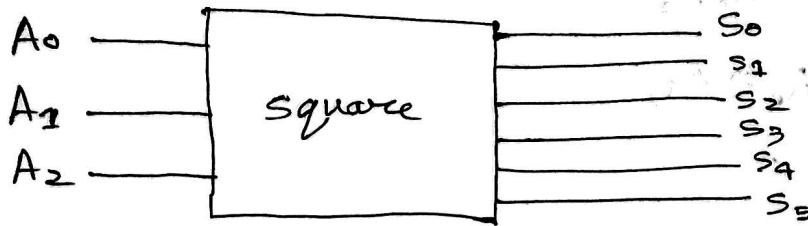
d) @

A B C D



4.2) Input - 3 bit number.

Output - Square of the input number.



Truth table

A_0	A_1	A_2	S_0	S_1	S_2	S_3	S_4	S_5
0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	1	1
0	1	0	0	0	0	1	0	0
0	1	1	0	0	1	0	0	1
1	0	0	0	1	0	0	0	0
1	0	1	0	1	1	0	0	1
1	1	0	1	0	0	1	0	0
1	1	1	1	1	0	0	0	1

$S_0, S_4 = 0, S_5 = A_2$

Kmap for S_0

$A_0 \backslash A_1 A_2$	00	01	11	10
0				
1			1	1

$S_0 = A_0 A_1$

Kmap for S_1

$A_0 \backslash A_1 A_2$	00	01	11	10
0				
1	1	1	1	

$S_1 = A_0 \bar{A}_1 + A_0 A_2$

Kmap for S_2

$A_0 \backslash A_1 A_2$	00	01	11	10
0			1	
1		1		

$S_2 = A_2 (A_0 \oplus A_1)$

Kmap for S_3

$A_0 \backslash A_1 A_2$	00	01	11	10
0				1
1				1

$S_3 = A_1 \bar{A}_2$

So, final Expression,

$$S_0 = A_0 A_1$$

$$S_1 = A_0 \bar{A}_1 + A_0 A_2$$

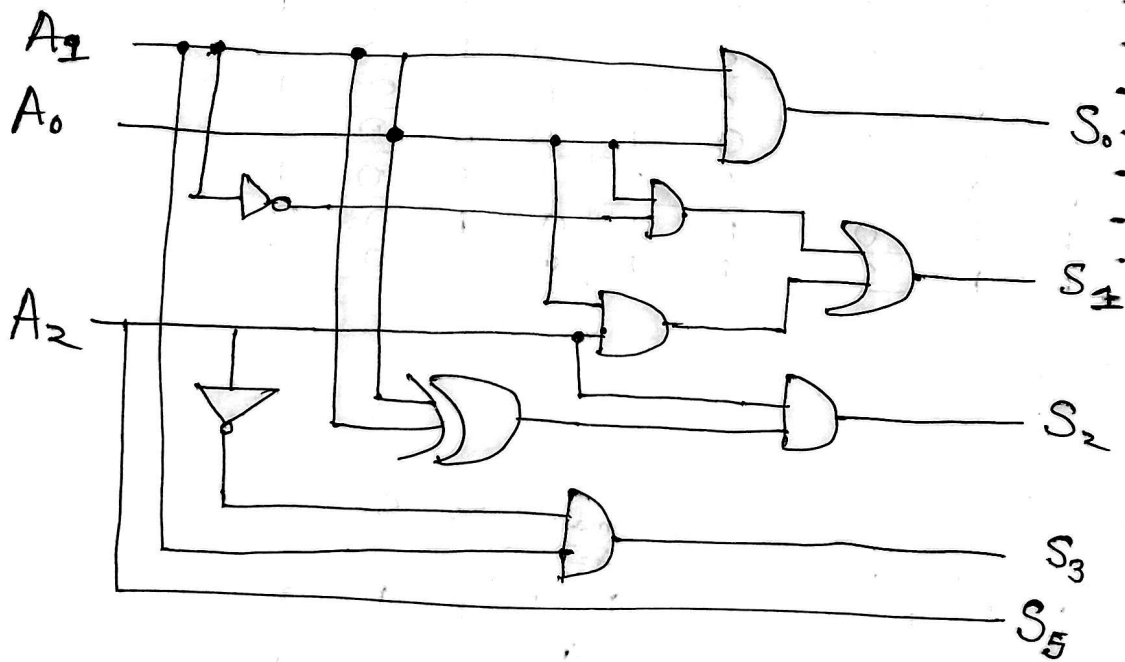
$$S_2 = A_2 (A_0 \oplus A_1)$$

$$S_3 = A_1 \bar{A}_2$$

$$S_4 = 0$$

$$S_5 = A_2$$

Circuit diagram



4.3)

(a)

$$\begin{array}{r}
 a_1 a_0 \\
 \times b_1 b_0 \\
 \hline
 b_0 a_1 \quad b_0 a_0 \\
 b_1 a_1 \quad b_1 a_0 \\
 \hline
 p_3 \quad p_2 \quad p_1 \quad p_0
 \end{array}$$

So, there will four output lines required.

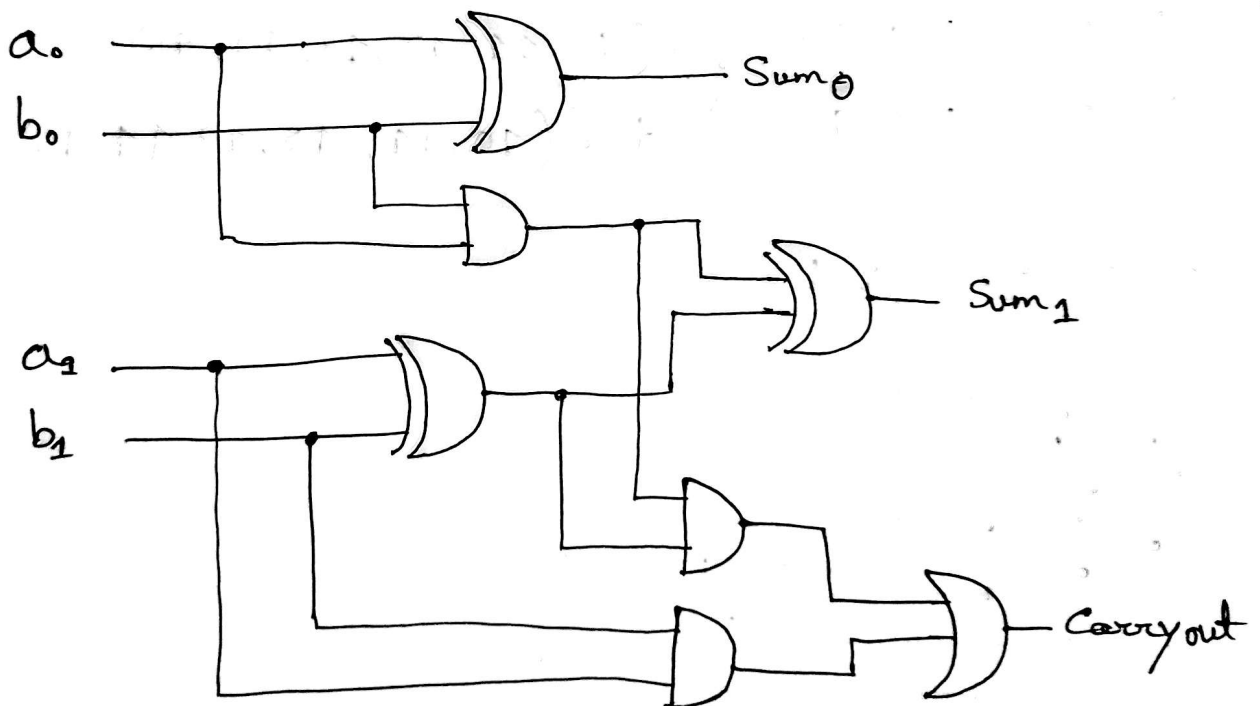
(b) $P_0 = b_0 a_0$

$P_1 = b_0 a_1 \oplus b_1 a_0$ (Carry will generate which will propagate to next)

$P_2 = b_1 a_1 \oplus ((b_0 a_1) \cdot (b_1 a_0))$

$P_3 = a_1 b_1 + ((b_0 a_1) \cdot (b_1 a_0))$ (Ans)

4.4)



4.5)

BCD input				Decimal	9's complement	output							
A	B	C	D	d	9-d	w	x	y	z	w	x	y	z
0	0	0	0	0	9	1	0	0	1	1	0	0	1
0	0	0	1	1	8	1	0	0	0	1	0	0	0
0	0	1	0	2	7	0	1	1	1	0	1	1	1
0	0	1	1	3	6	0	1	1	0	0	1	1	0
0	1	0	0	4	5	0	1	0	1	0	1	0	1
0	1	0	1	5	4	0	1	0	0	0	1	0	0
0	1	1	0	6	3	0	0	1	1	0	0	1	1
1	0	0	0	7	2	0	0	1	0	0	0	1	0
1	0	0	1	8	1	0	0	0	1	0	0	0	1
1	0	1	0	9	0	0	0	0	0	0	0	0	0

Boolean function:

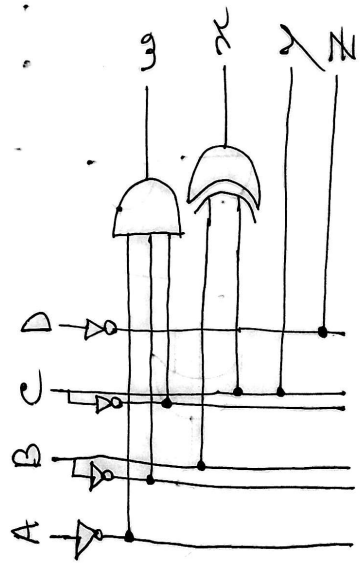
$$w = \Sigma (0, 1) + d(10, 11, 12, 13, 14, 15)$$

$$x = \Sigma (2, 3, 4, 5) + d(10, 11, 12, 13, 14, 15)$$

$$y = \Sigma (7, 3, 6, 7) + d(10, 11, 12, 13, 14, 15)$$

$$z = \Sigma (0, 2, 4, 6, 8) + d(10, 11, 12, 13, 14, 15)$$

Logic circuit:



4.6)

Binary number				2's complement number			
A	B	C	D	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	0	1	1	0
1	0	1	1	0	1	0	1
1	1	0	0	0	1	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	0	1	0
1	1	1	1	0	0	0	1

$$Y_3 = \sum m(1, 2, 3, 4, 5, 6, 7, 8)$$

$$Y_2 = \sum m(1, 2, 3, 4, 9, 10, 11, 12)$$

$$Y_1 = \sum m(1, 2, 5, 6, 9, 10, 13, 14)$$

$$Y_0 = \sum m(1, 3, 5, 7, 9, 11, 13, 15)$$

$$Y_3 = \bar{A}B + \bar{A}D + \bar{A}C + A\bar{B}\bar{C}\bar{D}$$

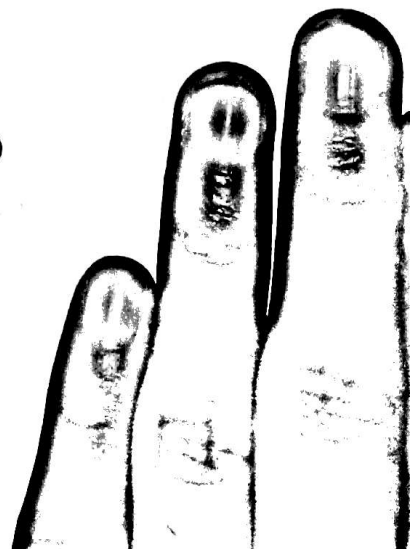
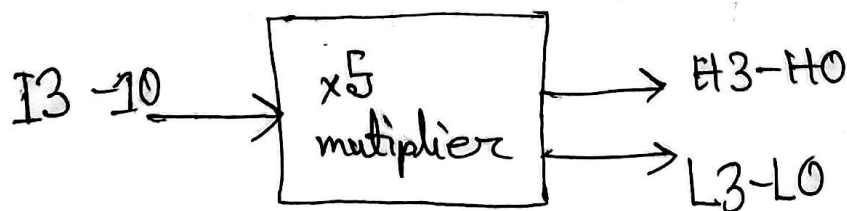
$$Y_2 = \bar{B}C + \bar{B}D + B\bar{C}\bar{D}$$

$$Y_1 = \bar{C}D + C\bar{D}$$

$$Y_0 = D$$

4.7) Each digit of the decimal number is coded separately in binary form into 4 bit binary number.

Since the circuit provides output that is 5 times the BCD digit, so output value can contain maximum of 2 decimal digits or 8 binary digits. Let I_3-I_0 be the 4 bit BCD digit input to the $\times 5$ multiplier and H_3-H_0 and L_3-L_0 be the 8 bit-BCD output as shown in the below.



the truth table for $\times 5$ multiplier -

BCD _{Input}	BCD _{Output}	T_3	T_2	T_1	T_0	H_3	H_2	H_1	H_0	L_3	L_2	L_1	L_0
0	00	0	0	0	0	0	0	0	0	0	0	0	0
1	05	0	0	0	1	0	0	0	0	0	0	0	0
2	10	0	0	1	0	0	0	0	0	0	1	0	1
3	15	0	0	1	1	0	0	0	1	0	0	0	0
4	20	0	1	0	0	0	0	1	0	0	1	0	1
5	25	0	1	0	1	0	0	1	0	0	0	0	0
6	30	0	1	1	0	0	0	1	1	0	0	0	0
7	35	0	1	1	1	0	0	1	1	0	1	0	1
8	40	1	0	0	0	0	1	0	0	0	0	0	0
9	45	1	0	0	1	0	1	0	0	0	1	0	0

we can ignore rest,

Kmap for H_3 , L_3 and L_1

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

$$\Rightarrow H_3 = L_3 = L_1 = 0$$

Similarly, we get, $H_2 = T_3$, $H_1 = T_2$, $H_0 = T_1$,
 $L_2 = L_0 = T_0$

So, the whole function can be implemented without a single logic gate.

4.8) The truth table below contains all six entries for which, $F = 1$. No others correspond to the unused bit combinations in the BCD code.

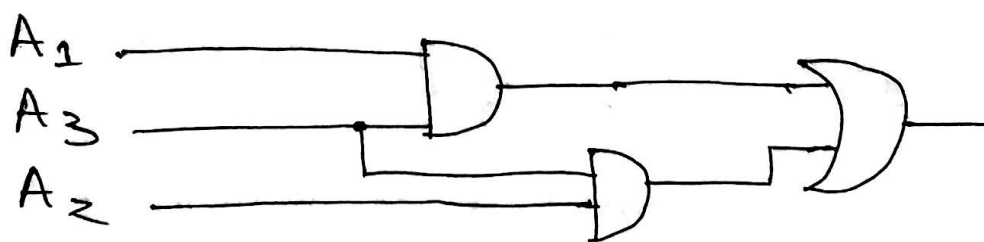
A_3	A_2	A_1	A_0	F
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Hence, from the table we have sum of product expression

$$F = A_3(\bar{A}_2 A_1 \bar{A}_0 + \bar{A}_2 A_1 A_0 + A_2 \bar{A}_1 \bar{A}_0 + A_2 \bar{A}_1 A_0 + A_2 A_1 \bar{A}_0 + A_2 A_1 A_0)$$

This can be simplified as -

$$F = A_3 A_2 + A_3 A_1 \quad \text{(Ans)}$$

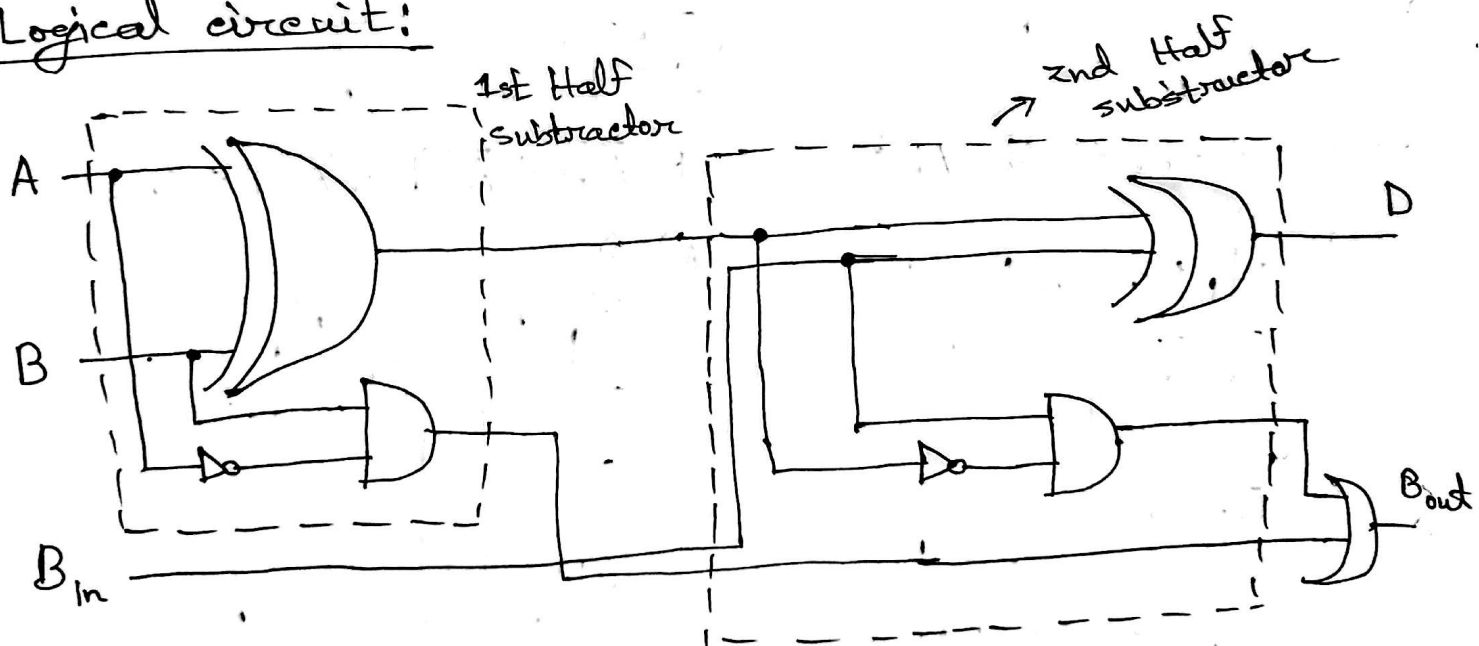


4.9) In full subtractor, Logical expression for difference, $D = A \oplus B \oplus B_{in}$

Logical expression for borrow -

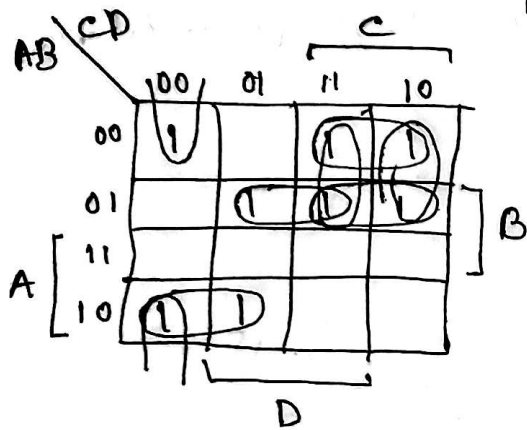
$$\begin{aligned} B_{out} &= \bar{A}B + \bar{A}B_{in} + BB_{in} \\ &= \bar{A}B + \bar{A}B_{in}(B + \bar{B}) + BB_{in}(A + \bar{A}) \\ &= \bar{A}B + \bar{A}BB_{in} + \bar{A}\bar{B}\bar{B}_{in} + ABB_{in} + \bar{A}BB_{in} \\ &= \bar{A}B + B_{in}(\bar{A}\bar{B} + AB) \\ &= \bar{A}B + B_{in}(A \oplus B) \end{aligned}$$

Logical circuit:

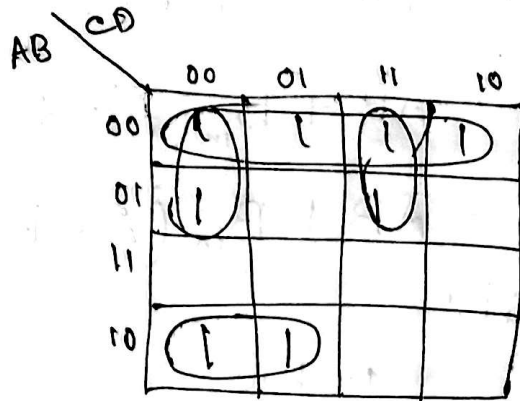


4.10)

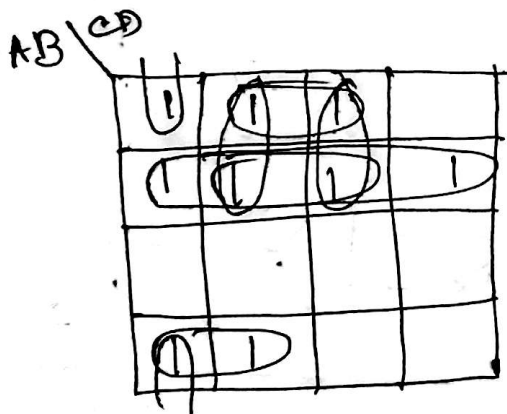
ABCD	a	b	c	d	e	f	g
0000	1	1	1	1	1	1	1
0001	0	1	1	0	0	0	0
0010	1	1	0	1	1	0	0
0011	1	1	1	1	0	0	1
0100	0	1	1	0	0	1	1
0101	1	0	1	1	0	1	1
0110	1	0	1	1	1	1	1
0111	1	1	1	0	0	1	1
1000	1	1	1	1	1	0	0
1001	1	1	1	1	0	1	1



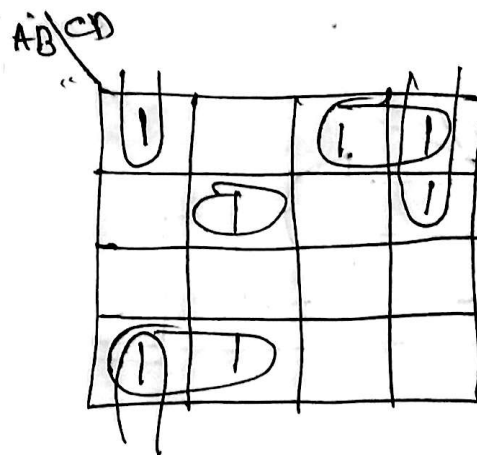
$$a = \bar{A}\bar{C} + \bar{A}BD + \bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}$$



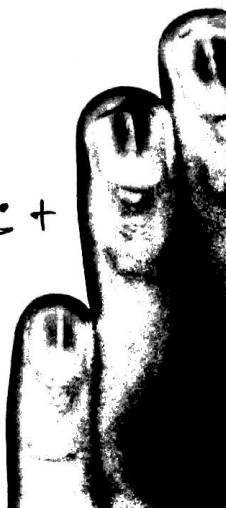
$$b = \bar{A}\bar{B} + \bar{A}\bar{C}\bar{D} + \bar{A}C\bar{D} + A\bar{B}\bar{C}$$

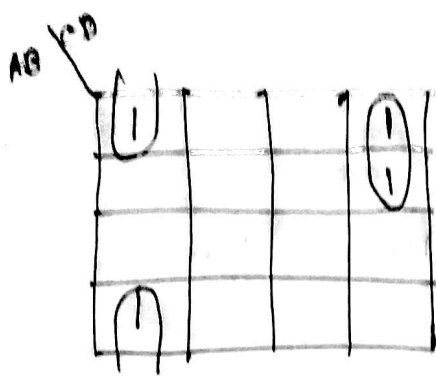


$$c = \bar{A}\bar{B} + \bar{A}\bar{D} + \bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}$$

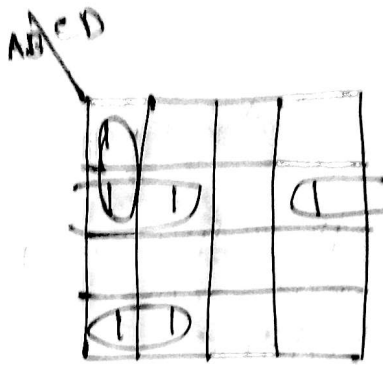


$$d = \bar{A}\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D$$

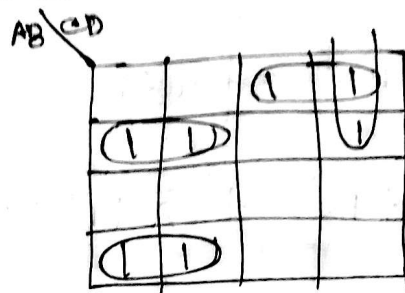




$$e = \bar{A}\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$



$$f = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{C}\bar{D} + \bar{A}B\bar{D} + \bar{A}\bar{B}\bar{C}$$



$$g = \bar{A}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

4.11) Here Boolean functions are F_1, F_2

$$\begin{aligned} F_1 &= \overline{(AB + BC + AC)} (A + B + C) + ABC \\ &= (\bar{A}\bar{B} \cdot \bar{B}\bar{C} \cdot \bar{A}\bar{C}) (A + B + C) + ABC \\ &= (\bar{A} + \bar{B}) (\bar{B} + \bar{C}) (\bar{A} + \bar{C}) (A + B + C) + ABC \\ &= (\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{B} + \bar{B}\bar{C}) (\bar{A} + \bar{C}) (A + B + C) + ABC \\ &= (\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}(1 + \bar{C})) (\bar{A} + \bar{C}) (A + B + C) + ABC \\ &= \{ \bar{B}(1 + \bar{A}) + \bar{A}\bar{C} \} (\bar{A} + \bar{C}) (A + B + C) + ABC \\ &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + ABC \end{aligned}$$

and, $F_2 = AB + AC + BC$

4.14) From the figure 4.15, we have

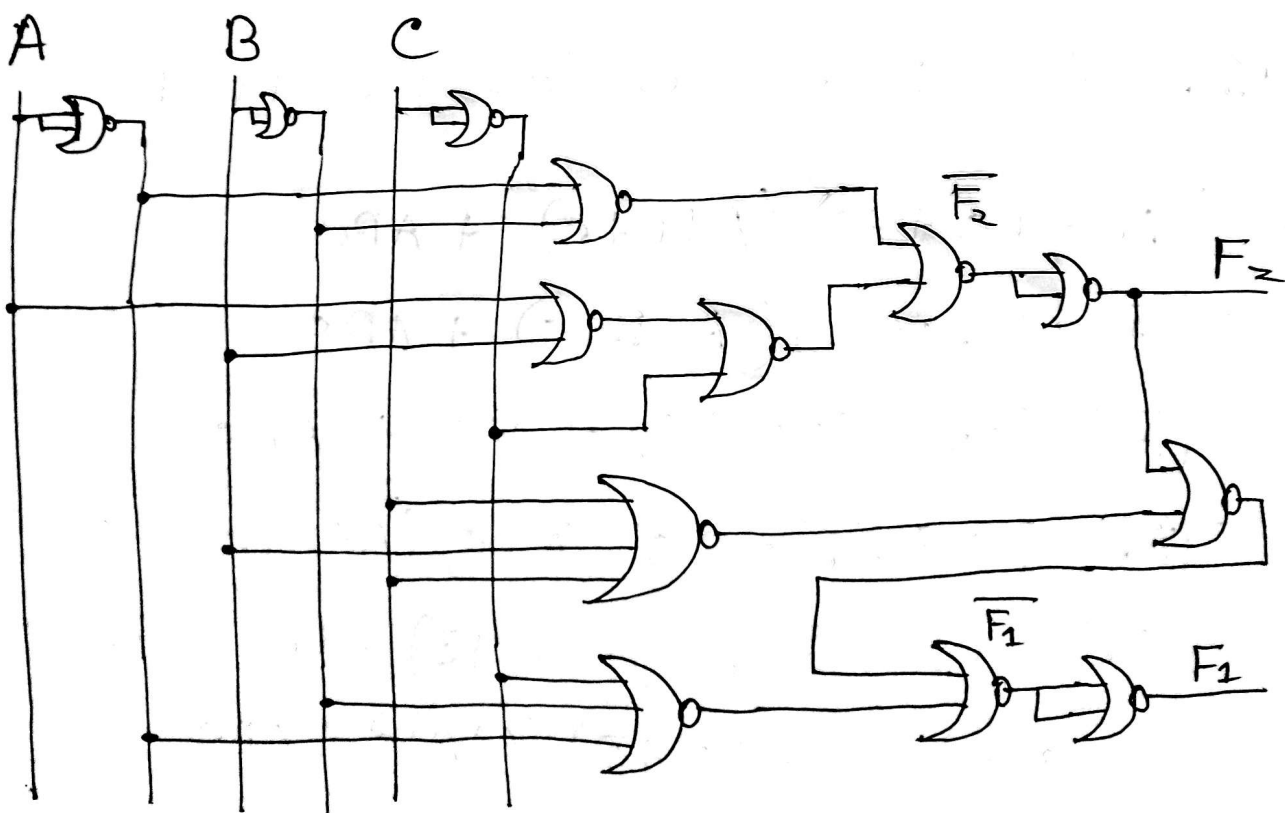
$$F_2 = AB + (A+B)C$$

$$F_1 = (A+B+C) \overline{F_2} + ABC$$

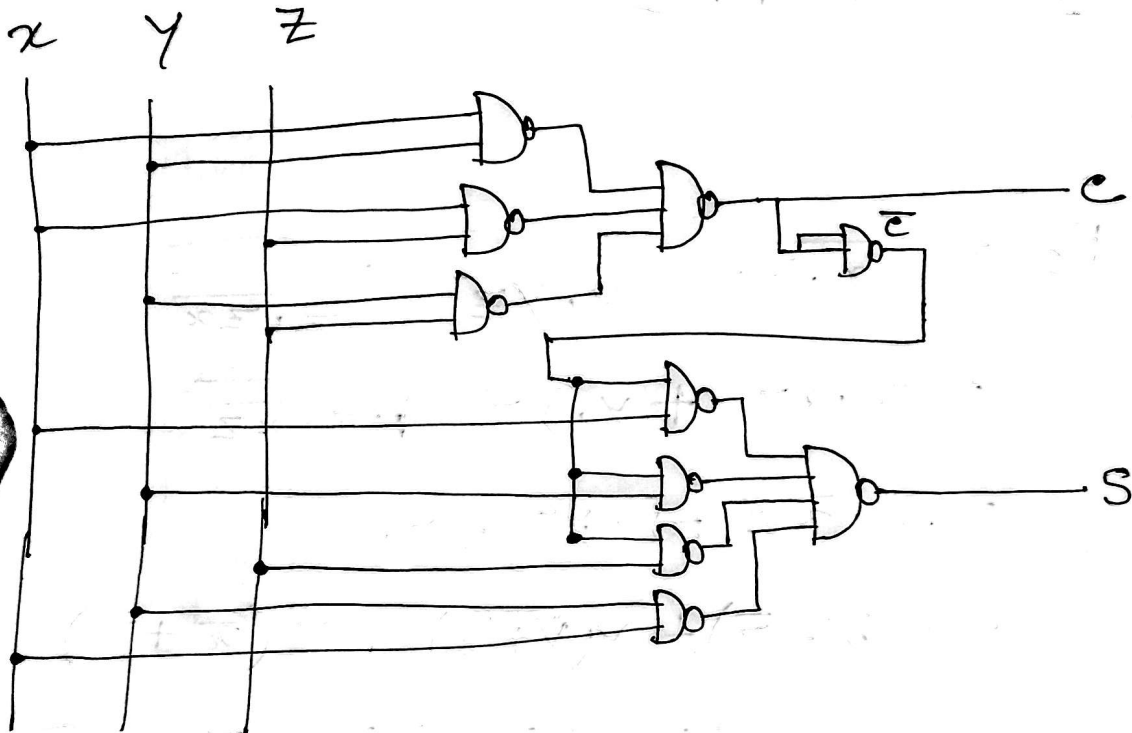
$$\therefore \overline{F_2} = \overline{AB + (A+B)C} = \left[\overline{(\overline{A} + \overline{B})} + \left\{ \overline{(A+B) + C} \right\} \right]$$

$$\overline{F_1} = \left[\left\{ \overline{(A+B+C) + F_2} \right\} + \overline{(\overline{A} + \overline{B} + \overline{C})} \right]$$

Now implementation:

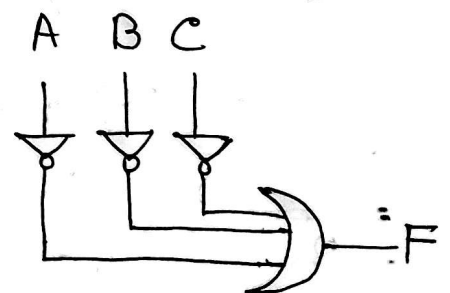


4.15) $S = \bar{c}(x+y+z) + xyz$
 $c = xy + xz + yz$



4.16) Boolean function for F,

$$\begin{aligned}
 F &= \overline{\overline{A+B+C} + \overline{A+B} + \overline{B}} \\
 &= \overline{\overline{A+B+C} + \overline{A+B} + \overline{B}} \\
 &= \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{A} \cdot \overline{B} + \overline{B} \\
 &= ABC + \overline{A}B + \overline{B} \\
 &= ABC + \overline{A} + \overline{B} [A + \overline{A}B = A + B] \\
 &= \overline{A} + \overline{B} + \overline{C}
 \end{aligned}$$



Logic circuit

4.17)

$$a) F_1 = \overline{x \cdot xz \cdot y \cdot yz \cdot z \cdot zx}$$

$$F_2 = \overline{xy \cdot xz}$$

4.18)

$$a) F_1 = \overline{x \cdot xz \cdot y \cdot yz \cdot z \cdot zx}$$

$$= \overline{x \cdot xz} + \overline{y \cdot yz} + \overline{z \cdot zx}$$

$$= x \cdot \overline{xz} + y \cdot \overline{yz} + z \cdot \overline{zx}$$

$$= x(\overline{x} + \overline{z}) + y(\overline{y} + \overline{z}) + z(\overline{x} + \overline{z})$$

$$= x\overline{x} + x\overline{z} + y\overline{y} + y\overline{z} + z\overline{x} + z\overline{z}$$

$$= x\overline{z} + y\overline{z} + z\overline{x}$$

$$F_2 = \overline{xy \cdot xz}$$

$$= xy + xz$$

x	y	z	$\overline{x}$	$\overline{z}$	F ₁	F ₂
0	0	0	1	1	0	0
0	0	1	1	0	1	0
0	1	0	1	1	1	0
0	1	1	1	0	1	0
1	0	0	0	1	1	0
1	0	1	0	0	0	1
1	1	0	0	1	1	1
1	1	1	0	0	0	1

4.18) b)

$$\begin{aligned}
 F &= \overline{\overline{A+B} + B + A + \overline{A+B} + A + \overline{A+B} + \overline{A+B} + B + C + A + \overline{A+B} + A+B + B + C + C} \\
 &= \overline{\overline{A+B} + B + A + \overline{A+B} + A + \overline{A+B} + \overline{A+B} + B + C + A + \overline{A+B} + \overline{A+B} + B + C + C} \\
 &= \overline{\overline{A+B} + B + A + \overline{A+B} + \overline{A+B} + B + C + A + \overline{A+B} + \overline{A+B} + B + C + C} \\
 &= \overline{\overline{A+B} + B + A + \overline{A+B} + \overline{A+B} + B + C + A + \overline{A+B} + \overline{A+B} + B + C + C} \\
 &= (\overline{B} (A+B) + \overline{A} (A+B) + \overline{C} (A+B) (B+\overline{A})) (C + \overline{C} (A+B) (B+\overline{A})) \quad \left[\begin{array}{l} \text{Absorption, distributive,} \\ \text{Idempotent, Involution} \\ \text{laws} \end{array} \right] \\
 &= \overline{B}AC + \overline{A}CB + \overline{C}AB + \overline{C} (B+\overline{A}) \overline{B} \overline{A} A + \overline{C} (B+\overline{A}) \overline{A} \overline{B} \\
 &= \overline{A} \overline{B} C + \overline{A} B C + A \overline{B} \overline{C} + \overline{C} \overline{B} \overline{A} B + \overline{C} \overline{B} \overline{A} \overline{A} \\
 &= \overline{B}AC + \overline{A} B C + \overline{C}AB + \overline{C} \overline{B} \overline{A} B + \overline{C} \overline{B} \overline{A} \\
 &= \overline{A} \overline{B} C + \overline{A} B C + A \overline{B} \overline{C} + 0 + \overline{A} \overline{B} \overline{C} \\
 &= \overline{A} \overline{B} \overline{C} + \overline{A} B C + A \overline{B} C + A \overline{B} \overline{C}
 \end{aligned}$$

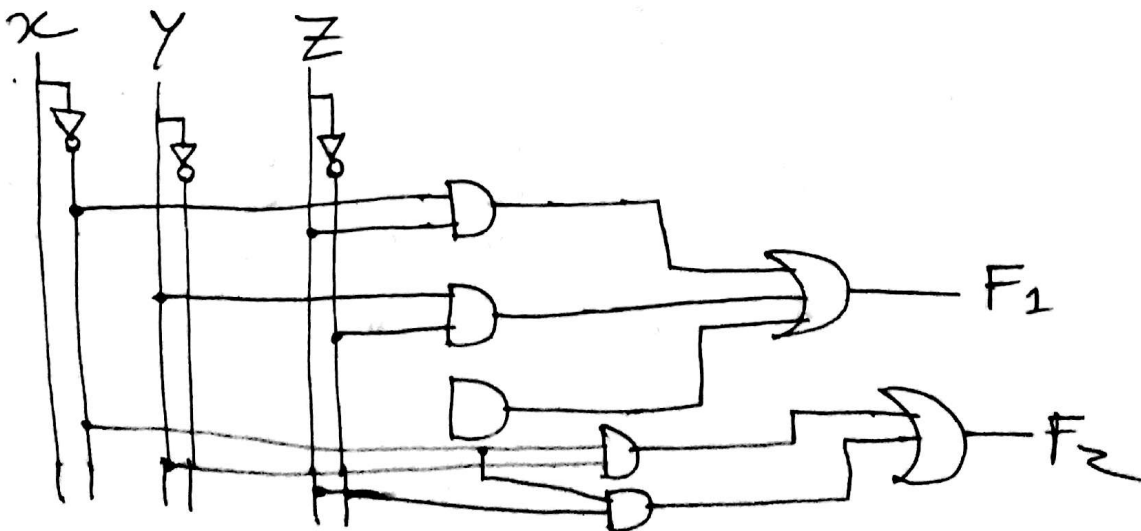
$$\therefore F = \overline{A}\overline{B}\overline{C} + \overline{A}BC + A\overline{B}C + ABC\overline{C}$$

Truth table

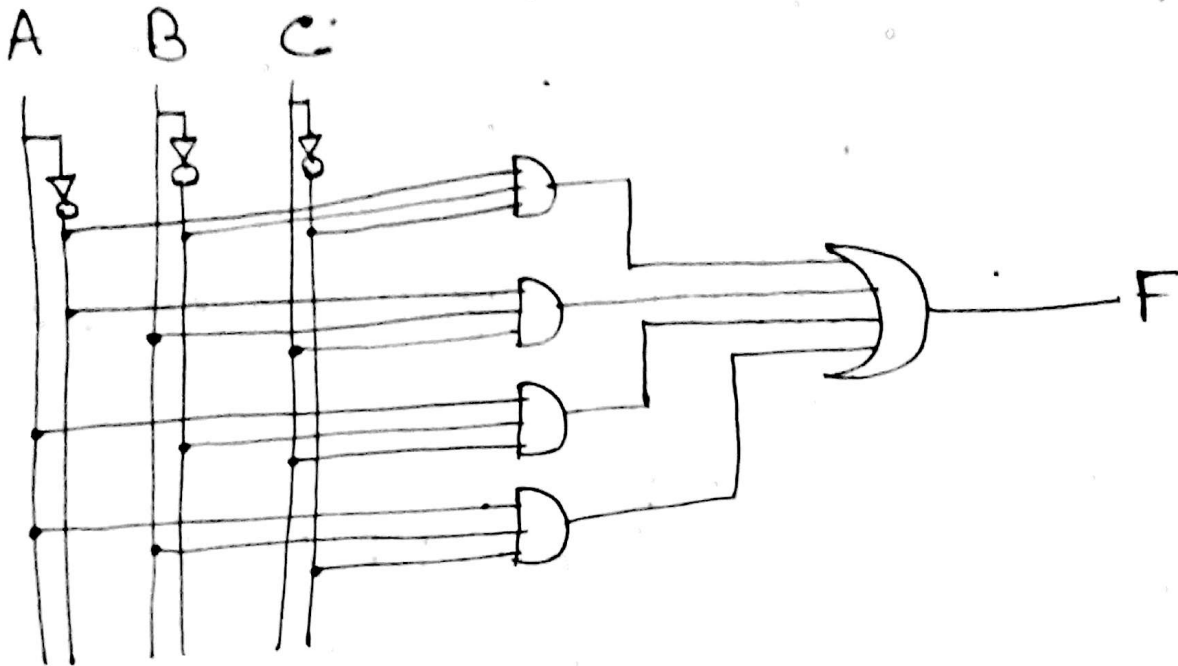
A	B	C	$\overline{A}$	$\overline{B}$	$\overline{C}$	$\overline{A}\overline{B}\overline{C}$	$\overline{A}BC$	$A\overline{B}C$	$ABC\overline{C}$	$\overline{A}\overline{B}\overline{C} + \overline{A}BC + A\overline{B}C + ABC\overline{C}$
0	0	0	1	1	1	1	0	0	0	1
0	0	1	1	1	0	0	0	0	0	0
0	1	0	1	0	1	0	0	0	0	0
0	1	1	1	0	0	0	1	0	0	1
1	0	0	0	1	1	0	0	0	0	0
1	0	1	0	1	0	0	0	1	0	1
1	1	0	0	0	1	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0

4.19) $F_1 = x\overline{z} + y\overline{z} + z\overline{x}$

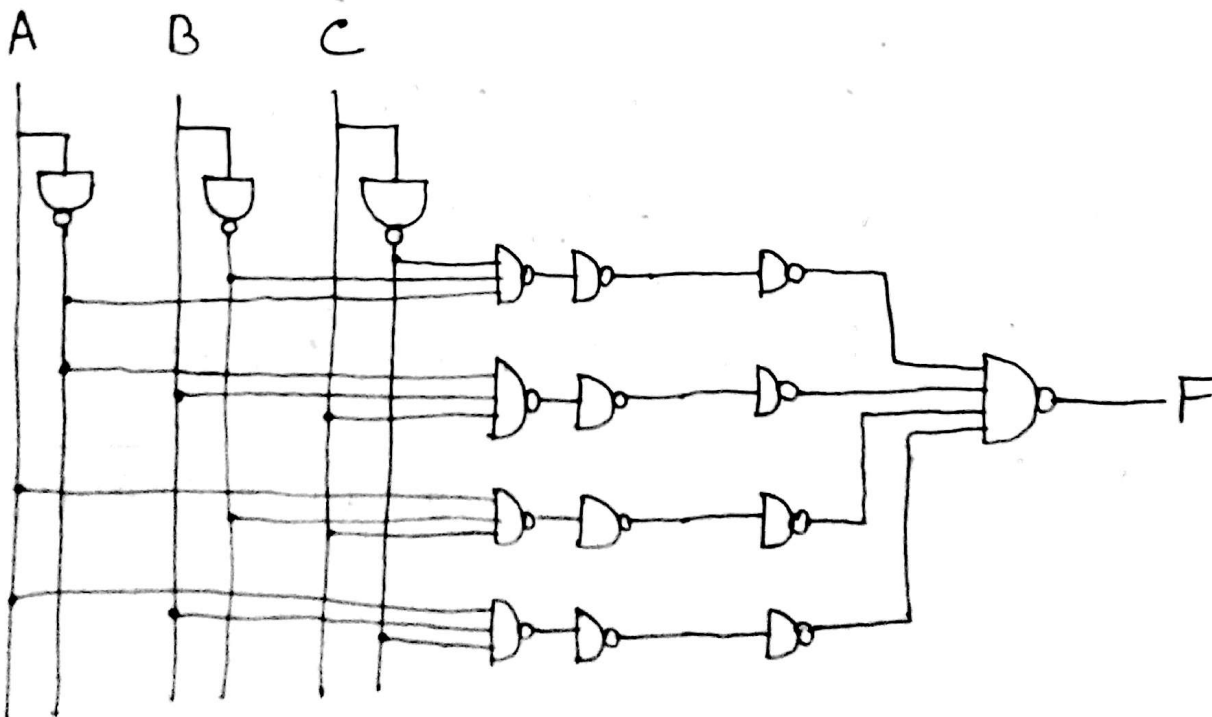
$$F_2 = xy + xz$$



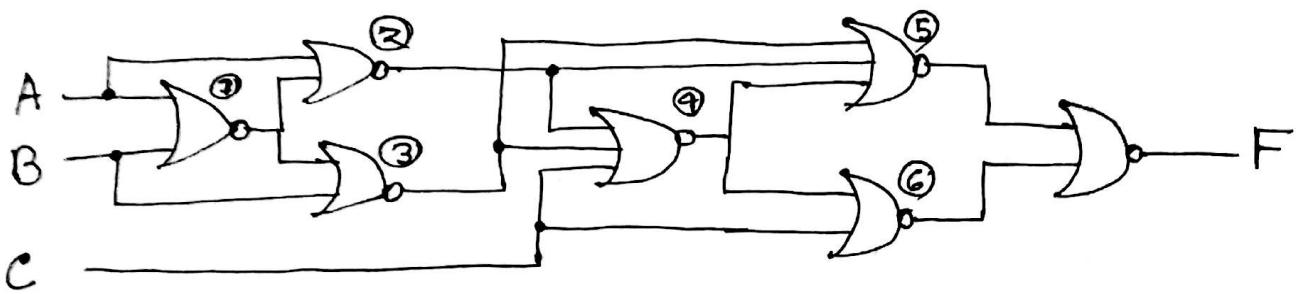
4.20) $F = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C + ABC\bar{C}$



4.21) logic diagram using NAND gates -
 $F = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C + ABC\bar{C}$



4.22 fig-4.21 (b) - expression -



Let,

$$1 = \overline{A+B} \quad \therefore 2 = \overline{A + \overline{A+B}} = \overline{A} \cdot \overline{\overline{A+B}} = \overline{A} (A+B) = \overline{A}B$$

$$\begin{aligned} \therefore 3 &= \overline{B + \overline{A+B}} \\ &= \overline{B} (A+B) \\ &= A\overline{B} \end{aligned}$$

$$\begin{aligned} \therefore 4 &= \overline{2+3+C} \\ &= \overline{\overline{A}B + A\overline{B} + C} \\ &= \overline{\overline{A}B + A\overline{B} + C} \\ &= \overline{\overline{A}B} \cdot \overline{A\overline{B}} \cdot \overline{C} \\ &= (\overline{\overline{A} + B})(\overline{A + \overline{B}}) \cdot \overline{C} \\ &= (AB + \overline{A}\overline{B})\overline{C} \end{aligned}$$

$$\therefore 5 = \overline{2+3+4} = \overline{\overline{A}B + A\overline{B} + \overline{C}(AB + \overline{A}\overline{B})}$$

$$\begin{aligned} &= \overline{\overline{A}B} \cdot \overline{A\overline{B}} \cdot \overline{\overline{C}(AB + \overline{A}\overline{B})} = (AB + \overline{A}\overline{B})(\overline{C} + \overline{(AB + \overline{A}\overline{B})}) \\ &= (AB + \overline{A}\overline{B})(\overline{C} + (\overline{AB} \cdot \overline{\overline{A}\overline{B}})) = (AB + \overline{A}\overline{B})(\overline{C} + (\overline{A+B})(A+B)) \\ &= (AB + \overline{A}\overline{B})(\overline{C} + \overline{A}B + A\overline{B}) = ABC + \overline{A}\overline{B}C = C(\overline{A \oplus B}) \end{aligned}$$

$$\begin{aligned}
 \therefore G &= \overline{A+C} = \overline{(AB+\bar{A}\bar{B})\bar{C} + C} = \overline{AB\bar{C} + \bar{A}\bar{B}\bar{C} + C} \\
 &= (\bar{A}+\bar{B}+C)(A+B+C)\bar{C} = \bar{C}(\bar{A}B + \bar{A}C + A\bar{B} + \bar{B}C + AC + BC + \bar{A}B) \\
 &= \bar{C}(\bar{A}B + A\bar{B}) = \bar{C}(A \oplus B)
 \end{aligned}$$

$$\begin{aligned}
 \therefore F &= \overline{G+C} = \overline{C(A \oplus B) + C(A \oplus B)} \\
 &= \overline{A \oplus B \oplus C}
 \end{aligned}$$

So, we can say that circuit in fig-4.21(b) is a exclusive-OR.

